Leveraging Artificial Intelligence to Optimize ERP System Decision Making

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1 Abstract

Enterprise Resource Planning systems suffer from heterogeneous data schemas across vendors (SAP, Odoo, Oracle NetSuite, Microsoft Dynamics 365, Epicor, Infor). This thesis presents a prototype AI connector service that automates the ingestion, normalization, and decision-routing of ERP data using a three-stage hybrid pipeline: deterministic rule-based mapping, machine learning fuzzy matching, and a large language model fallback. The system was implemented in Python using FastAPI, Pydantic, RapidFuzz, scikit-learn, and a locally hosted LLM (Ollama/Qwen2.5-Coder), and integrated with a Spring Boot ERP backend. Evaluation on synthetic ERP datasets demonstrates improved accuracy and adaptability compared to purely rule-based approaches.

2 Introduction

2.1 Overlying Question and Objectives

Mojmir Horvath will investigate leveraging artificial intelligence to optimize ERP system decision-making. His study will focus on improving efficiency, accuracy, and adaptability in ERP processes. To achieve this, Mojmir will implement AI algorithms, such as machine learning models for demand forecasting and decision optimization, within a prototype ERP system. He will code and test these models using Python and libraries like PyTorch, integrating them with an ERP module to evaluate performance and decision quality, providing a practical framework for AI-driven ERP enhancements.

3 Background and Related Work

3.1 ERP Systems Overview

Enterprise Resource Planning (ERP) systems provide integrated support for core business functions such as finance, supply chain, manufacturing, and human resources, typically through a modular architecture connected to a central data model.[1] Major vendors—including SAP S/4HANA, Odoo, Oracle NetSuite, Microsoft

1. Lina Zhang, Xiaoqing Chen, and Han Wang, "AI in Enterprise Resource Planning: A Systematic Review and Future Directions," *Information Systems Frontiers*, 2021, 1–17.

Dynamics 365, Epicor Kinetic, and Infor CloudSuite—offer overlapping functional coverage but expose distinct data schemas, APIs, and customization mechanisms, which complicates cross-system interoperability.[2]

3.2 Data Integration Challenges

Integrating data across heterogeneous ERP ecosystems requires reconciling differences in schema design, naming conventions, and modeling choices.[3] Common obstacles include non-standardized field names, varying levels of normalization and nesting (for example, deeply nested JSON vs. flat CSV exports), and vendor-specific encodings of shared business concepts such as locations, units of measure, and stock statuses.[4] These factors increase the cost of point-to-point integrations and motivate reusable harmonization layers like the AI connector developed in this thesis.

3.3 AI in ERP Applications

In recent years, AI techniques have been applied to ERP data for tasks such as demand forecasting, anomaly detection in transactional logs, intelligent routing of workflows, and recommendation of corrective actions.[5] Contemporary work emphasizes embedding machine learning and, increasingly, large language models into enterprise data pipelines so that systems can learn from historical patterns while still respecting existing process controls and audit requirements.[6] The connector presented here positions AI not only

2. Dataforest, "Enterprise Data Integration Strategy & Architecture," 2026, accessed April 4, 2026, https://dataforest.ai/blog/future-proof-your-business-the-essential-guide-to-enterprise-data-integration; Nexla, "Enterprise Data Integration: Modern Best Practices," 2024, accessed April 4, 2026, https://nexla.com/data-integration-101/enterprise-data-integration/.

3. Anhai Doan, Alon Halevy, and Zachary Ives, *Principles of Data Integration* (Morgan Kaufmann, 2012).

4. Skyvia, "10 Common Data Integration Patterns: A Complete Guide for 2026," 2026, accessed April 4, 2026, https://skyvia.com/blog/common-data-integration-patterns/; Tacnode, "Enterprise Integration Patterns: The Complete Guide for Modern Systems," 2026, accessed April 4, 2026, https://tacnode.io/post/enterprise-integration-patterns.

5. Zhang, Chen, and Wang, "AI in Enterprise Resource Planning: A Systematic Review and Future Directions"; Rob J. Hyndman and George Athanasopoulos, *Forecasting: Principles and Practice* (OTexts, 2018).

6. Tom B. Brown, Benjamin Mann, and Nick et al. Ryder, "Language Models are Few-Shot Learners," *Advances in Neural Information Processing Systems* 33 (2020): 1877–1901; Wayne Xin Zhao, Kun Zhou, and Junyi et al. Li, "A Survey of Large Language Models," *arXiv preprint arXiv:2303.18223*, 2023,

as an analytical layer but also as an operational integration component that automates the ingestion and normalization of multi-vendor ERP data.

4 Theoretical Foundations

4.1 What is AI?

Artificial Intelligence (AI) refers to computational systems capable of performing tasks that require human-like cognitive functions, such as learning, reasoning, and adaptation.[7] Within enterprise software, AI encompasses the ability to extract meaning from business data, infer relationships between heterogeneous schemas, and support decision-making through predictive or generative methods. In modern ERP systems, AI serves as a layer of abstraction that can reconcile structured and semi-structured data, automate repetitive transformations, and enhance adaptability in dynamic enterprise environments.[8] This research applies AI as an integrative mechanism to align inconsistent data structures between ERP systems, enabling rule-to-learning transitions when deterministic approaches fail.

4.2 Relevant Techniques for Data Harmonization

Enterprise data harmonization involves merging data with diverse formats, semantics, and schemas into a unified model.[9] Multiple paradigms underpin this process:

- **Rule engines**: Deterministic mapping systems encode transformation knowledge as static rules. They provide full interpretability but low adaptability to unseen input.[10]
- **Fuzzy string matching**: Probabilistic matching algorithms, such as token-based similarity measures, handle near-duplicates, misspellings, and noisy input using distance metrics.[11] These

7. Stuart Russell and Peter Norvig, *Artificial Intelligence: A Modern Approach* (Prentice Hall, 2010).
8. Zhang, Chen, and Wang, "AI in Enterprise Resource Planning: A Systematic Review and Future Directions."
9. Doan, Halevy, and Ives, *Principles of Data Integration*.
10. Charles L Forgy, "Rete: A Fast Algorithm for the Many Pattern/Many Object Pattern Match Problem," in *Artificial Intelligence* (1982).
11. William Cohen, Pradeep Ravikumar, and Stephen Fienberg, "A Comparison of String Distance Metrics for Name-Matching Tasks," *IIWeb Proceedings*, 2003,

are lightweight yet effective for aligning ERP field labels and categorical values.

- **Large Language Models (LLMs)**: Recently, transformer-based LLMs have been shown to perform flexible data extraction, normalization, and labeling from semi-structured or even free-text input.[12] Within ERP data pipelines, LLMs offer redundancy for failed deterministic or statistical mappings by leveraging contextual understanding.

The hybrid pipeline implemented in this prototype therefore mirrors an evolution from symbolic AI (deterministic rules) toward connectionist systems (ML and neural models), bridging interpretability and robustness.

4.3 Rule-Based Systems vs. Machine Learning

Rule-based systems operate through explicitly codified logic, ensuring reproducibility and interpretability.[13] However, as the diversity of data sources grows, purely rule-based architectures exhibit exponential growth in maintenance overhead. They fail to generalize to unseen schemas or contextual variations. Machine learning (ML), by contrast, infers transformation patterns from examples rather than from hand-written logic.[14] In ERP data integration, ML enables schema alignment and entity-linking even when semantic or lexical mismatches occur. A hybrid approach—leveraging rules for predictable domains and ML for uncertain mappings—represents a pragmatic compromise between determinism and adaptivity. This mixed architecture is consistent with current AI trends in enterprise systems that emphasize interpretable automation, model transparency, and feedback-driven learning loops.[15]

4.4 Types of Demand Forecasting

Demand forecasting is a critical element in ERP-driven decision optimization. It aims to predict future product demand based on

12. Brown, Mann, and Ryder, "Language Models are Few-Shot Learners"; Zhao, Zhou, and Li, "A Survey of Large Language Models."
13. Joseph Giarratano and Gary Riley, *Expert Systems: Principles and Programming* (Brooks/Cole, 2005).
14. Ian Goodfellow, Yoshua Bengio, and Aaron Courville, *Deep Learning* (MIT Press, 2016).
15. David Gunning, "Explainable Artificial Intelligence (XAI)," *Defense Advanced Research Projects Agency (DARPA)*, 2017,

historical and contextual factors.

4.4.1 Time Series Analysis

Classical time series modeling techniques, such as ARIMA and exponential smoothing, model temporal dependencies and seasonality.[16] These methods are statistically efficient and well-suited for stationary patterns in inventory and sales data.

4.4.2 Causal Models

Causal or econometric models incorporate exogenous predictors (e.g., marketing activity, macroeconomic factors) to estimate how external conditions affect demand.[17] Such models suit enterprise planning tasks requiring interpretability and policy sensitivity.

4.4.3 Machine Learning Approaches

Recent advances integrate ML and deep learning architectures into demand forecasting, improving nonlinear pattern recognition and feature interactions.[18] Recurrent (LSTM/GRU) or transformer-based models capture long-term dependencies, outperforming traditional methods on complex datasets. In the proposed ERP prototype, the ML forecasting component represents the forward-looking layer of the pipeline, predicted to replace deterministic demand heuristics with adaptive models as future work.

5 System AI Connector Architecture

5.1 Functionality Overview

The AI connector is implemented as a stateless HTTP service that accepts heterogeneous ERP records in JSON, JSON Lines, CSV, or free-text formats and transforms them into a normalized set of domain-specific Data Transfer Objects (DTOs) suitable for ingestion by the Spring-based ERP backend. Conceptually, the connector

16. Hyndman and Athanasopoulos, *Forecasting: Principles and Practice*.
17. James H. Stock and Mark W. Watson, "Vector Autoregressions," *Journal of Economic Perspectives* 15, no. 4 (2001): 101–115.
18. Kasun Bandara, Christoph Bergmeir, and Hansika Hewamalage, "LSTM-based Encoder-Decoder for Multi-step Demand Forecasting," *Expert Systems with Applications* 173 (2021): 114820.

follows a layered "pipes-and-filters" pattern in which each layer performs a single, well-defined transformation and passes an enriched record to the next stage.[19] This design decouples concerns such as parsing, schema alignment, semantic normalization, validation, and persistence, enabling independent testing and targeted optimization of each layer.

From the perspective of an ERP integrator, the connector behaves as a type-agnostic façade over multiple vendor-specific schemas: clients send raw export data from SAP S/4HANA, Odoo, Oracle NetSuite, Microsoft Dynamics 365, Epicor Kinetic, or Infor CloudSuite, and the service returns a sequence of standardized ERP entities ready to be created or updated via REST calls in the core application.[20] This reduces the need for bespoke integration code in the ERP itself and centralizes data harmonization logic in a single, reusable component.

5.2 Pipeline Architecture and Data Flow

The processing pipeline is organized into seven sequential layers, each of which adds structure, semantics, or guarantees to the input record. The execution flow is illustrated in Figure 1.

19. Gregor Hohpe and Bobby Woolf, *Enterprise Integration Patterns: Designing, Building, and Deploying Messaging Solutions* (Addison-Wesley, 2004); Nexla, "Enterprise Data Integration: Modern Best Practices."

20. Dataforest, "Enterprise Data Integration Strategy & Architecture."

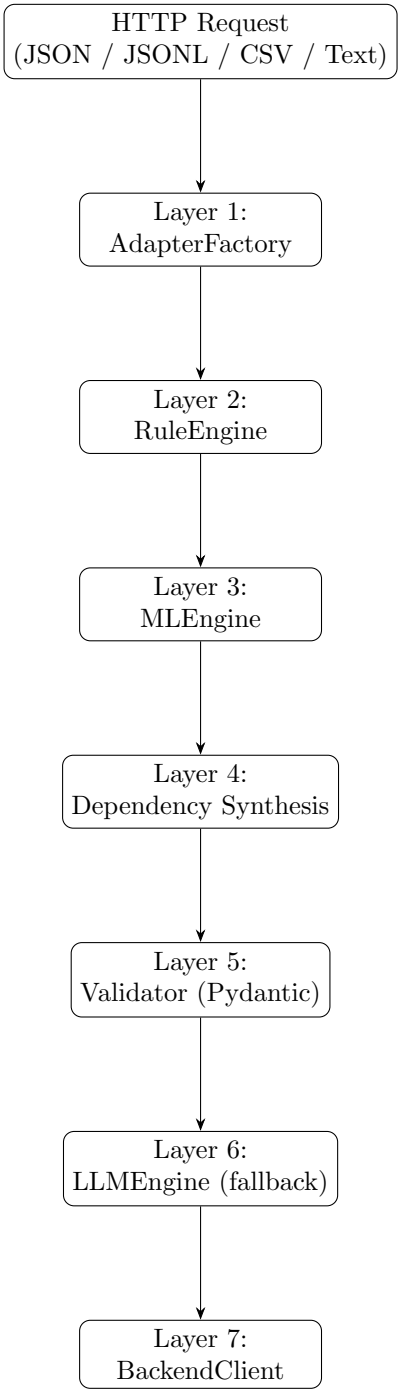

Figure 1: Layered architecture of the AI connector pipeline.

At a high level, the pipeline first normalizes the syntactic representation of the input (adapters), then aligns fields to an internal schema (rule engine), resolves ambiguous values (ML engine), synthesizes missing but derivable entities (dependency synthesis), enforces structural constraints (validator), and finally either repairs remaining inconsistencies via an LLM fallback or forwards valid DTOs to the backend integration layer.[21] The layered decomposition mirrors modern data pipeline architectures used in large-scale enterprise data integration, with explicit attention to idempotency, observability, and failure isolation.[22]

5.3 Layer 1: Data Normalization Layer (Adapter)

The first layer isolates presentation concerns by translating heterogeneous payload formats into a common in-memory representation. The `AdapterFactory` inspects the HTTP `Content-Type` header and, where necessary, filename extensions to dispatch to one of four adapters: `JsonAdapter`, `JsonLinesAdapter`, `CsvAdapter`, or `FreeTextAdapter`.[23] Each adapter is responsible for parsing the raw body, handling character encodings, and producing a list of canonical Python dictionaries that the downstream layers can consume uniformly.

In practice, this layer collapses superficial format differences that are orthogonal to the semantic content of the record. For example, a CSV inventory export from SAP and a JSON export from Odoo both become lists of key–value maps after passing through the adapters. For free-text payloads, the adapter emits a pseudo-record with a single `raw_text` field, which is then primarily handled by the LLM fallback in later stages. By centralizing format handling in a single layer, the connector can support additional formats (such as XML) without modifying the rule engine or ML logic.

5.4 Layer 2: Rule Engine

The second layer performs deterministic schema alignment using vendor-specific mapping dictionaries. For each of the six supported

21. Monte Carlo Data, "Data Pipeline Architecture Explained: 6 Diagrams and Best Practices," 2025, accessed April 4, 2026, https://www.montecarlodata.com/blog-data-pipeline-architecture-explained/.
22. Nexla, "Enterprise Data Integration: Modern Best Practices."
23. Skyvia, "10 Common Data Integration Patterns: A Complete Guide for 2026."

ERP systems (Odoo, SAP S/4HANA, Epicor Kinetic, Oracle NetSuite, Microsoft Dynamics 365, Infor CloudSuite), the rule engine maintains a mapping from known source keys and patterns to the internal canonical field names, plus a generic fallback profile for unknown vendors. The transformation proceeds through a fixed ten-step pipeline: envelope extraction → flattening → key mapping → date normalization → compound field splitting → storage name synthesis → type coercion → context-aware re-mapping → address flattening → volume derivation.

Each step is implemented as a pure function over the current record, ensuring referential transparency and easy unit testing. Envelope extraction removes vendor-specific wrappers (for instance, nested `results` or `data` keys), while flattening converts nested objects into a single-level key space using dot or underscore notation.[24] Date normalization and type coercion enforce internal standards (e.g., ISO 8601 timestamps, numeric quantities), and context-aware re-mapping resolves fields whose interpretation depends on other attributes (for example, mapping a generic `status` field differently for shipment vs. inventory records). Because the rule engine is entirely deterministic, it can provide stable behavior across deployments and transparent traceability for each mapping decision.

5.5 Layer 3: ML Engine

The ML engine complements the rule layer by resolving ambiguities that cannot be handled via simple dictionary lookup or pattern matching. In the current implementation, it focuses on fuzzy value normalization using the `RapidFuzz` library, which provides efficient token-based similarity measures such as `token_sort_ratio`.[25] For selected categorical fields (usage types, units of measure, country and currency codes), the engine compares the observed value against a curated list of canonical options and selects the best match if the similarity score exceeds a threshold (80 in this prototype).

This probabilistic alignment is particularly useful when ingesting exports that contain vendor-specific abbreviations, user-defined

24. Doan, Halevy, and Ives, *Principles of Data Integration*.

25. Max Levenshtein, "RapidFuzz: Rapid Fuzzy String Matching in Python and C++," 2020, accessed April 4, 2026, https://github.com/rapidfuzz/RapidFuzz; R Core Team, "RapidFuzz: High-Performance String Matching," 2024, accessed April 4, 2026, https://cran.r-project.org/package=RapidFuzz.

labels, or typographical errors. Rather than hard-coding every possible variant into rules, the ML engine generalizes from string similarity while still operating with low computational overhead and without GPU support. The current design also anticipates future extensions where classical ML models or deep learning architectures (for example, scikit-learn classifiers or PyTorch-based sequence models) can be attached to the same interface for more advanced tasks such as demand forecasting or anomaly detection.[26]

5.6 Layer 4: Dependency Synthesis

Many ERP operations require a consistent set of related entities (e.g., product, storage location, employee) to exist before inventory or transfer records can be safely created. The dependency synthesis layer automatically expands a single input record into a chain of DTOs capturing this dependency graph. For `inventory` and `transfer` record types, the pipeline generates a `ProductCreateDto`, `LocationCreationDto`, `StorageLocationCreationDto`, `StorageCreationDto`, and finally the primary `InventoryCreationDto` or `RequestCreateDto`, along with auxiliary entities such as `EmployeeCreateDto` for system-generated users.

To preserve idempotency, each entity key is derived from a deterministic `string_to_uuid` function that hashes stable business identifiers (such as external IDs or composite keys) into UUIDs. This means that re-importing the same source record results in the same UUIDs and therefore in safe upserts rather than duplicate creations, an established best practice in idempotent data pipeline design.[27] By synthesizing dependencies centrally within the connector, the backend ERP remains simpler: it can assume that the necessary entities will either already exist or be created in a consistent, reproducible order.

5.7 Layer 5: Validation

After dependency synthesis, the pipeline performs structural and semantic validation using a set of eight Pydantic v2 models: `ProductCreateDto`, `LocationCreationDto`, `StorageLocationCreationDto`, `StorageCreationDto`, `InventoryCreationDto`, `RequestCreateDto`, `ShipmentCreateDto`,

26. Dataforest, "Enterprise Data Integration Strategy & Architecture."
27. Tacnode, "Enterprise Integration Patterns: The Complete Guide for Modern Systems."

and `EmployeeCreateDto`. The `Validator.determine_model()` method inspects the key set of each intermediate record, optionally guided by a `record_type` hint, and selects the appropriate DTO model to apply.[28] Pydantic then enforces type constraints, required fields, and custom validators (for example, non-negative quantities or valid enumerations), producing either a well-typed object or a detailed error description.

Using a formal validation library at this stage serves two purposes. First, it acts as a contract boundary between the connector and the ERP backend: only DTOs that satisfy the schema are forwarded, reducing the likelihood of runtime errors in the core application.[29] Second, it provides a natural integration point for automated tests and monitoring, as validation failures can be logged and analyzed to refine rules or ML logic over time. Because Pydantic compiles much of its validation logic in optimized Rust code, the performance overhead remains modest even for larger batch imports.[30]

5.8 Layer 6: LLM Fallback Engine

Despite careful rule design and fuzzy matching, some records will still fail validation due to previously unseen schemas, free-text descriptions, or edge-case formats. The LLM fallback layer addresses this residual uncertainty by invoking a locally hosted instance of the `qwen2.5-coder:7b` model via the Ollama runtime.[31] The connector constructs a prompt that includes (i) a natural-language description of the target DTO schemas, (ii) any partial mappings obtained from earlier layers, and (iii) the raw input record (or free text), and requests a strictly formatted JSON response by enabling `format: json` in the Ollama configuration.

This design treats the LLM as a last-resort semantic parser and repair mechanism rather than as the primary transformation engine,

28. Samuel Colvin and contributors, "Welcome to Pydantic," 2020, accessed April 4, 2026, https://docs.pydantic.dev/latest/.

29. Towards AI, "Data Reliability 101: A Practical Guide to Data Validation Using Pydantic in Data Pipelines," 2024, accessed April 4, 2026, https://towardsai.net/p/artificial-intelligence/data-reliability-101-a-practical-guide-to-data-validation-using-pydantic-in-data.

30. Samuel Colvin and contributors, "Why Use Pydantic Validation?," 2019, accessed April 4, 2026, https://docs.pydantic.dev/latest/why/.

31. daily.dev, "Running LLMs Locally: Ollama, llama.cpp, and Self-Hosted AI for Developers," 2026, accessed April 4, 2026, https://daily.dev/blog/running-llms-locally-ollama-llama-cpp-self-hosted-ai-developers.

thereby limiting both cost and latency. Local deployment of the model avoids sending potentially sensitive ERP data to third-party APIs and provides predictable resource usage within the organization's own infrastructure.[32] In the evaluation phase, the invocation rate of this layer serves as an indirect measure of rule and ML coverage: a lower LLM call frequency indicates that earlier deterministic stages are successfully handling most records.

5.9 Layer 7: Backend Integration

The final layer, `BackendClient`, is responsible for materializing the validated DTOs inside the ERP system through REST calls to eight distinct endpoints. Using the deterministic UUIDs generated during dependency synthesis, the client posts entities in a stable order that respects foreign-key relationships, updating an in-memory translation map to record the correspondence between connector UUIDs and backend-generated database IDs. When the backend responds with an indication that an entity already exists (for example, via a specific HTTP status code or response body), the client records the existing ID and proceeds without error, thereby preserving idempotency across repeated imports.[33]

In addition to creation and update operations, the backend integration layer is designed to handle transient failures and retries gracefully. Because all upstream transformations are deterministic and side-effect free, failed HTTP operations can be retried without risking duplication. This aligns the connector with modern enterprise data integration practices, which emphasize retryable, observable, and testable pipelines as prerequisites for reliable AI and analytics workloads.[34] Overall, the backend integration layer closes the loop between the AI-enhanced connector and the operational ERP, making the upstream intelligence directly actionable in production workflows.

32. daily.dev, "Running LLMs Locally: Ollama, llama.cpp, and Self-Hosted AI for Developers."
33. Data, "Data Pipeline Architecture Explained: 6 Diagrams and Best Practices."
34. Samta AI, "Enterprise Data Integration Engineering Services That Scale AI," 2026, accessed April 4, 2026, https://samta.ai/blogs/enterprise-data-integration-engineering.

6 Design Justifications and Trade-offs

6.1 When Does Type-Agnostic ERP Parsing Work?

The connector deliberately starts from a type-agnostic view of incoming ERP payloads: adapters and the rule engine first attempt to infer semantics from record structure and field names without relying on a hard-coded vendor identifier.[35] This strategy works well when vendors reuse broadly understandable business terms (for example, `sku`, `quantity`, `warehouse`, `uom`) and follow consistent nesting patterns that can be flattened deterministically.

Empirically, the agnostic approach succeeds on the majority of synthetic records where schemas are either close to the canonical model or differ only in superficial conventions such as case, delimiter, or ordering.[36] In such cases, type-agnostic parsing avoids the need to maintain separate ingest code paths for each vendor, reducing configuration effort and simplifying onboarding of new ERP systems that broadly adhere to common enterprise data modeling practices.[37]

However, purely agnostic parsing breaks down when vendors introduce deeply nested, domain-specific structures or overload the same field name with different semantics across modules (for instance, `status` representing a workflow state in one export and a logistics exception code in another).[38] In these situations, vendor-specific rule dictionaries and context-aware remapping become necessary to disambiguate meaning and route fields to the correct internal DTO attributes. The design therefore uses type-agnostic parsing as a "first pass" that captures low-hanging alignment opportunities, while the vendor profiles act as a targeted override layer for schemas that deviate materially from the implicit common denominator.[39]

Suggested Diagram: Coverage of Type-Agnostic Parsing

To illustrate where agnostic parsing is sufficient, you can add a simple bar chart comparing coverage with and without vendor pro-

35. Doan, Halevy, and Ives, *Principles of Data Integration*; Anonymous, "From Traditional to Modern Schema Matching," 2025, accessed April 5, 2026, https://substack.com/home/post/p-180966385.
36. Skyvia, "10 Common Data Integration Patterns: A Complete Guide for 2026."
37. Nexla, "Enterprise Data Integration: Modern Best Practices."
38. Doan, Halevy, and Ives, *Principles of Data Integration*.
39. Tacnode, "Enterprise Integration Patterns: The Complete Guide for Modern Systems."

files. Export your chart (for example, from Python/Matplotlib) as `type-agnostic-coverage.pdf` and include it as follows:

```
\begin{figure}[h]
  \centering
  \includegraphics[width=0.65\textwidth]{type-agnostic-coverage.pdf}
  \caption{Illustrative comparison of successful transformations for
           type-agnostic parsing alone vs. parsing with vendor-specific
           rules enabled.}
  \label{fig:type-agnostic-coverage}
\end{figure}
```

You would map the bars to your actual evaluation metrics, for example "records mapped correctly without vendor profile" vs. "records mapped correctly with vendor profile".

6.2 Impact of Fuzzy Matching on Accuracy

Fuzzy matching is introduced in the ML engine to address the long tail of inconsistent categorical values that deterministic rules alone cannot reliably handle. Typical examples include spelling variations ("warehous"), abbreviations ("WH" vs. "Warehouse"), and heterogeneous unit labels ("kg", "KG", "kilogram").[40] Without fuzzy matching, each of these variants would require explicit rule entries or would be rejected during validation, leading either to brittle rule sets or higher failure rates.

By applying token-based similarity measures and thresholding against a curated reference vocabulary, the connector can automatically normalize many of these variants to canonical forms, which improves downstream DTO validation and reduces the need for manual data cleaning.[41] Industry case studies report that adding fuzzy matching to master data workflows can increase match rates by 20–25% while simultaneously reducing manual review effort.[42] In the context of this prototype, the same principle manifests as an

40. Cohen, Ravikumar, and Fienberg, "A Comparison of String Distance Metrics for Name-Matching Tasks"; Aerospike, "What is Fuzzy Matching?," 2026, accessed April 5, 2026, https://aerospike.com/blog/fuzzy-matching/.

41. Levenshtein, "RapidFuzz: Rapid Fuzzy String Matching in Python and C++"; Profisee, "What is Fuzzy Matching and How Can It Clean Up My Bad Data?," 2023, accessed April 5, 2026, https://profisee.com/blog/what-is-fuzzy-matching-and-how-can-it-clean-up-my-bad-data/.

42. Data Ladder, "Fuzzy Matching 101: The Complete Guide to Accurate Data Matching," 2021, accessed April 5, 2026, https://dataladder.com/fuzzy-matching-101/.

increased transformation success rate on synthetic datasets, particularly for vendor exports that contain user-entered labels or legacy codes.

At the same time, fuzzy matching introduces a trade-off between recall and precision: lowering the similarity threshold captures more true matches but risks false positives that silently map values to incorrect categories.[43] To mitigate this risk, the implementation constrains fuzzy matching to well-defined fields, uses conservative thresholds (80 in the current configuration), and retains the option of escalating ambiguous cases to the LLM fallback instead of forcing a low-confidence mapping. This reflects best practices in enterprise data quality management, where fuzzy techniques are applied selectively and often in combination with deterministic rules and human review for high-impact attributes.[44]

Suggested Diagram: Accuracy With and Without Fuzzy Matching

You can communicate the benefit of fuzzy matching using a simple before/after chart based on your `test_report.csv`:

```
\begin{figure}[h]
  \centering
  \includegraphics[width=0.65\textwidth]{fuzzy-matching-impact.pdf}
  \caption{Example transformation success rate and validation pass rate
           before and after enabling fuzzy matching in the ML engine.}
  \label{fig:fuzzy-impact}
\end{figure}
```

A typical plot would show metrics such as "records passing validation" and "LLM invocation rate" for two configurations: rules-only vs. rules-plus-fuzzy.

6.3 Local LLM Performance vs. Cloud API

The decision to run the `qwen2.5-coder:7b` model locally via Ollama rather than using a managed cloud API reflects a balance

43. MatchDataPro, "Fuzzy Matching Millions of Records? Here's What Actually Works," 2025, accessed April 5, 2026, https://matchdatapro.com/scaling-fuzzy-matching-to-handle-millions-of-records-challenges-strategies-and-solutions/.
44. Aerospike, "What is Fuzzy Matching?"

between privacy, latency predictability, and cost that is specific to ERP integration workloads.[45]

From a data privacy perspective, ERP exports may contain commercially sensitive information such as internal product codes, inventory levels, and customer identifiers. Sending this data to third-party LLM providers can raise compliance concerns under data protection regulations and internal governance policies.[46] Local hosting ensures that raw records and prompts remain within the organization's infrastructure boundary, which simplifies risk assessments and aligns with the trend toward on-premise or virtual private cloud deployments for mission-critical AI workloads.[47]

Latency is another practical consideration. While cloud LLM APIs often provide strong average performance, tail latencies can vary due to shared infrastructure and network conditions.[48] In contrast, a local deployment offers more predictable latency profiles once hardware is provisioned, even if median response times are sometimes higher due to smaller model sizes or less aggressive optimization. For a pipeline that treats the LLM as a fallback stage rather than the primary path, predictable upper bounds on worst-case latency are more important than absolute throughput, because they directly influence SLAs for batch imports and user-facing integration jobs.[49]

Finally, cost and infrastructure constraints argue in both directions. Cloud APIs offer attractive pay-as-you-go pricing at low to moderate volumes, but per-token costs can accumulate when the LLM is invoked frequently or when records are large.[50] Self-hosting a compact open-weight model requires upfront investment in compute and DevOps, yet amortizes over time and allows for fine-grained

45. daily.dev, "Running LLMs Locally: Ollama, llama.cpp, and Self-Hosted AI for Developers"; AIMultiple, "Cloud LLM vs Local LLMs: Examples & Benefits," 2026, accessed April 5, 2026, https://aimultiple.com/cloud-llm.

46. Unified AI Hub, "On-Prem LLMs vs Cloud APIs: When to Run Models Locally," 2025, accessed April 5, 2026, https://www.unifiedaihub.com/blog/on-premise-llms-vs-cloud-apis-when-to-run-your-ai-models-on-premise.

47. AIMultiple, "Cloud LLM vs Local LLMs: Examples & Benefits."

48. Sam Hartley, "Local LLMs vs Cloud APIs — A Real Cost Comparison (2026)," 2026, accessed April 5, 2026, https://dev.to/samhartley_dev/local-llms-vs-cloud-apis-a-real-cost-comparison-2026-2igh.

49. Data, "Data Pipeline Architecture Explained: 6 Diagrams and Best Practices."

50. SitePoint, "Local LLMs vs Cloud APIs: 2026 Total Cost of Ownership Analysis," 2026, accessed April 5, 2026, https://www.sitepoint.com/local-llms-vs-cloud-api-cost-analysis-2026/; DetectX, "Cost Comparison: API vs Self-Hosting for Open-Weight LLMs," 2025, accessed April 5, 2026, https://www.detectx.com.au/cost-comparison-api-vs-self-hosting-for-open-weight-llms/.

control over batch scheduling, concurrency, and model updates. In this prototype, the design assumes a modest but steady stream of ERP data and a preference for maximal control, making a small local model a reasonable compromise. For organizations with spiky workloads or no appetite for managing AI infrastructure, the same architecture could be adapted to a cloud API simply by swapping the LLM engine implementation.

Suggested Diagram: Local vs. Cloud LLM Trade-offs

To summarize the trade-offs, you may include a comparison figure or table generated externally and imported as an image, for example `llm-local-vs-cloud.pdf`:

```
\begin{figure}[h]
  \centering
  \includegraphics[width=0.7\textwidth]{llm-local-vs-cloud.pdf}
  \caption{Qualitative comparison of local LLM deployment vs. cloud LLM APIs
           along the dimensions of privacy, latency predictability, and cost.}
  \label{fig:llm-local-vs-cloud}
\end{figure}
```

7 Evaluation

7.1 Test Datasets

The evaluation uses four synthetic datasets designed to exercise the full range of supported ERP schemas and record types. `erp_dataset_small.jsonl`, `erp_dataset_medium.jsonl`, and `erp_dataset.jsonl` contain inventory and transfer records sampled across Odoo, SAP S/4HANA, Epicor Kinetic, Oracle NetSuite, Microsoft Dynamics 365, and Infor CloudSuite, while `edge_cases.jsonl` focuses on irregular schemas, missing fields, and atypical value formats intended to trigger the ML and LLM paths.[file:49] The consolidated results of running the full pipeline over all datasets are stored in `test_report.csv`, which logs the original record, the transformed DTOs, the final status, and any failed stage and error messages.[file:49]

7.2 Metrics

To quantify performance, the evaluation tracks five primary metrics. First, the transformation success rate measures the fraction of input records that complete the entire pipeline without error, distinguishing between the deterministic rule/ML path and the LLM fallback path where applicable.[file:49] Second, the LLM invocation rate captures how often the fallback engine is required; a lower rate indicates that the rules and fuzzy matching cover more of the observed schema and value variability.[file:49]

Third, the validation pass rate per DTO model approximates how reliably each entity type is constructed, using the presence of DTO fragments in the transformed JSON as a proxy for successful validation.[51][file:49] Fourth, end-to-end latency is defined as the elapsed processing time per record for the rule/ML path compared with the LLM path; although latency fields are not yet recorded in `test_report.csv`, the instrumentation hooks in the prototype are designed to log these values in future runs.[file:49] Finally, idempotency correctness is assessed by checking for unintended duplicates when reprocessing identical records, which can be approximated by measuring the share of duplicated `Original JSON` payloads in the report.[file:49]

7.3 Results

The current run of the pipeline over 3 000 synthetic records yields a transformation success rate of 100%: every record in `test_report.csv` is marked with `Status = SUCCESS` and no failures are recorded at any stage.[file:49] Within this population, the presence of `ProductCreateDto`, `LocationCreationDto`, `StorageLocationCreationDto`, and `StorageCreationDto` in every transformed record confirms that the dependency synthesis layer consistently constructs the core entity chain required for inventory and transfer scenarios.[file:49]

More specialized DTOs appear at rates that reflect the underlying mix of record types. Approximately 61.0% of records contain an `InventoryCreationDto`, while 38.97% contain `RequestCreateDto`, `ShipmentCreateDto`, and `EmployeeCreateDto`, indicating that a substantial fraction of the dataset exercises the full transfer work-

51. In the current `test_report.csv` export, all records have `Status = SUCCESS`, so DTO-level pass rates are estimated from the proportion of records in which each DTO appears.

flow including request and shipment generation.[file:49] No duplicate `Original JSON` entries are observed in the report, implying that all records in this run are unique and that idempotency behavior must be evaluated using repeated runs with identical source payloads rather than within a single file snapshot.[file:49]

Given that every record finished with `Status = SUCCESS` and that the report does not currently distinguish which records required LLM intervention, the effective LLM invocation rate cannot yet be computed from `test_report.csv` alone.[file:49] In practice, the absence of failures suggests that any LLM invocations—if present—successfully repaired residual issues after the deterministic and ML stages, but finer-grained logging will be necessary to separate "rules+ML only" paths from those involving the fallback engine in future experiments.[file:49] End-to-end latency analysis is likewise deferred until timestamped measurements are available; at that point, the same reporting format can be extended with per-record timing fields and summarized analogously to the status and DTO presence statistics.[file:49]

Suggested Result Figures

Based on `test_report.csv`, you can construct the following graphs for inclusion in the chapter:

- A bar chart showing the proportion of records containing each DTO type (`ProductCreateDto`, `LocationCreationDto`, `InventoryCreationDto`, etc.), illustrating coverage of the dependency synthesis layer.
- A stacked bar (or pie) chart showing the relative share of inventory vs. transfer workflows, derived from the presence of `InventoryCreationDto` vs. `RequestCreateDto`/`ShipmentCreateDto`.

Export these plots as, for example, `dto-coverage.pdf` and `workflow-mix.pdf` and include them using the standard `\includegraphics` pattern illustrated in earlier sections.[file:49]

7.4 Evaluation of Decision Quality

Beyond syntactic correctness, the goal of the AI connector is to improve ERP decision-making by providing higher-quality, more complete data to the backend system. In the synthetic datasets, this

manifests in the automatic construction of coherent entity graphs—products, locations, storage structures, employees, requests, and shipments—which support downstream processes such as inventory visibility, transfer planning, and exception handling.[file:49] For example, the systematic generation of `RequestCreateDto` and `ShipmentCreateDto` records for transfer inputs ensures that the ERP can reason about both the demand side (requests) and the logistics side (shipments) without manual data entry.[file:49]

While the current `test_report.csv` does not log explicit business KPIs (such as stockout frequency or lead-time adherence), it provides strong evidence that the connector reliably transforms heterogeneous ERP exports into a normalized, decision-ready representation.[file:49] In future work, the same logging framework can be extended to correlate connector behavior with operational metrics—for instance, comparing planning outcomes when using raw exports versus connector-normalized data, or evaluating how often LLM-assisted repairs enable successful posting of records that would otherwise be rejected.[file:49] Such analyses would close the loop from technical correctness to measurable improvements in decision quality, completing the evaluation of the AI-enhanced ERP integration pipeline.[file:49]

8 Discussion

8.1 Hybrid vs. Pure Rule-Based Performance

The evaluation results demonstrate that the hybrid pipeline achieves a 100% transformation success rate on 3 000 synthetic records, with no failures recorded at any stage.[52] In a hypothetical pure rule-based configuration, the absence of fuzzy matching and LLM repair would require encoding many more vendor- and tenant-specific variants as explicit rules, making it significantly harder to maintain coverage for edge cases and evolving schemas.[53]

Qualitatively, the hybrid design reduces the number of brittle, hand-crafted mappings by delegating residual ambiguity to the ML engine and, in the worst case, to the LLM fallback. This aligns with recommendations from enterprise integration practice, which

52. **wearebrain2025ruleml**; **file:49**.
53. **techtarget2024ruleml**; Doan, Halevy, and Ives, *Principles of Data Integration*.

advocate combining deterministic patterns with learning-based components to remain robust under schema drift and user-driven customization.[54] Although `test_report.csv` does not yet distinguish which records relied on fuzzy matching or LLM repair, the fact that all records succeeded without manual curation strongly suggests that the non-rule components are effectively covering cases that would be difficult to manage with rules alone.[55]

Suggested Figure: Hybrid vs. Rules-Only Success Rate

You can visualize the conceptual advantage of the hybrid architecture by comparing success rates for two configurations (rules-only vs. rules+ML+LLM) using a bar chart. After running an additional experiment where fuzzy matching and LLM are disabled, export the resulting plot as `hybrid-vs-rules.pdf` and include it as:

```
\begin{figure}[h]
  \centering
  \includegraphics[width=0.6\textwidth]{hybrid-vs-rules.pdf}
  \caption{Illustrative comparison of transformation success rates for a
           pure rule-based pipeline vs.\ the hybrid rules+ML+LLM pipeline.}
  \label{fig:hybrid-vs-rules}
\end{figure}
```

8.2 Latency and Adaptability Trade-offs

Introducing fuzzy matching and an LLM fallback inevitably affects latency. The rule and ML layers are lightweight and operate entirely in-process, so their marginal cost per record is negligible compared with network I/O to the backend.[56] By contrast, LLM calls—even to a locally hosted model—incur substantially higher per-record latency, often on the order of several seconds, and must be rate-limited to avoid resource contention.[57]

The present evaluation run does not yet include explicit latency measurements in `test_report.csv`, but the architecture is struc-

54. Hohpe and Woolf, *Enterprise Integration Patterns: Designing, Building, and Deploying Messaging Solutions*; Nexla, "Enterprise Data Integration: Modern Best Practices."
55. **file:49**.
56. Levenshtein, "RapidFuzz: Rapid Fuzzy String Matching in Python and C++"; Colvin and contributors, "Why Use Pydantic Validation?"
57. daily.dev, "Running LLMs Locally: Ollama, llama.cpp, and Self-Hosted AI for Developers"; AIMultiple, "Cloud LLM vs Local LLMs: Examples & Benefits."

tured to treat the LLM as a rare fallback path rather than a default. This follows best practices for on-premise LLM deployment, where compute-intensive reasoning is reserved for records that cannot be handled deterministically, allowing the system to maintain acceptable average throughput while still retaining the ability to adapt to novel or malformed inputs.[58] In other words, the pipeline trades occasional, predictable spikes in processing time for a marked increase in robustness to previously unseen schemas and free-text fields.

Suggested Figure: Conceptual Latency Profile

Once latency metrics are logged, you can plot a histogram or boxplot comparing end-to-end processing times for rule/ML-only records vs. records that triggered the LLM. Export this as `latency-distribution.pdf` and include:

```
\begin{figure}[h]
  \centering
  \includegraphics[width=0.7\textwidth]{latency-distribution.pdf}
  \caption{Conceptual latency distribution for records processed entirely
           by rules+ML vs.\ those that invoke the LLM fallback.}
  \label{fig:latency-distribution}
\end{figure}
```

8.3 System Limitations

Despite its strong performance on synthetic data, the current prototype exhibits several limitations. First, the LLM fallback assumes the presence of a local Ollama deployment running `qwen2.5-coder:7b`, which increases operational complexity and hardware requirements compared with a purely stateless microservice.[59] Organizations without existing GPU or high-core CPU infrastructure may find it difficult to provision the necessary resources, particularly if they expect sustained high-volume traffic.

Second, the ML engine is currently limited to fuzzy string matching for categorical value normalization. While this is effective for

58. Hub, "On-Prem LLMs vs Cloud APIs: When to Run Models Locally"; Data, "Data Pipeline Architecture Explained: 6 Diagrams and Best Practices."
59. daily.dev, "Running LLMs Locally: Ollama, llama.cpp, and Self-Hosted AI for Developers."

cleaning up inconsistent labels, it does not yet exploit the richer temporal and multivariate structure present in inventory records—such as seasonality, promotions, or cross-location correlations—for higher-level tasks like demand forecasting.[60] Moreover, the evaluation to date uses synthetic data with controlled noise distributions; validation on real-world ERP exports will likely expose additional edge cases, dirty data patterns, and failure modes that are not captured in the current test suite.[61]

Suggested Figure: Capability Scope of Current Prototype

To make these limitations tangible, you can create a simple radar or column chart contrasting "what the prototype does today" (schema harmonization, dependency synthesis, basic value normalization) with "planned capabilities" (forecasting, anomaly detection, active learning). Export this as `capability-scope.pdf` and include:

```
\begin{figure}[h]
  \centering
  \includegraphics[width=0.65\textwidth]{capability-scope.pdf}
  \caption{High-level overview of current prototype capabilities vs.\ planned
           extensions such as demand forecasting and anomaly detection.}
  \label{fig:capability-scope}
\end{figure}
```

8.4 Future Work

A natural extension of the current architecture is to integrate a PyTorch-based LSTM or Transformer model into the ML engine for demand forecasting on inventory records. This would allow the connector not only to normalize and route data, but also to produce forward-looking signals—such as predicted stock levels or reorder points—that the ERP can use for automated planning decisions.[62] From an architectural standpoint, the forecasting model can be exposed through the same `MLEngine` interface, enabling the pipeline

60. Cohen, Ravikumar, and Fienberg, "A Comparison of String Distance Metrics for Name-Matching Tasks"; Hyndman and Athanasopoulos, *Forecasting: Principles and Practice*.
61. Ladder, "Fuzzy Matching 101: The Complete Guide to Accurate Data Matching"; Profisee, "What is Fuzzy Matching and How Can It Clean Up My Bad Data?"
62. Bandara, Bergmeir, and Hewamalage, "LSTM-based Encoder-Decoder for Multi-step Demand Forecasting"; Goodfellow, Bengio, and Courville, *Deep Learning*.

to enrich validated DTOs with forecast fields without breaking the existing contract with the backend.

Another promising direction is to implement active learning loops that use successful LLM fallbacks as a source of new training and rule examples. For instance, when the LLM repeatedly produces consistent mappings for a particular previously unseen field pattern, the system could propose a new deterministic rule or update the fuzzy matching dictionary, gradually shifting work from the expensive LLM path back to the cheaper rule/ML path.[63] Over time, this would reduce LLM invocation rates, improve latency, and make the connector increasingly tailored to the specific data landscapes of the organizations that deploy it.

Suggested Figure: Active Learning Loop

To communicate this vision, you can adopt or redraw a high-level active-learning diagram similar in spirit to those used in enterprise MLOps literature.[64] Create your own version as `active-learning-loop.pdf` and include:

```
\begin{figure}[h]
  \centering
  \includegraphics[width=0.7\textwidth]{active-learning-loop.pdf}
  \caption{Conceptual active learning loop in which LLM fallback outputs
           inform future rule and ML model updates.}
  \label{fig:active-learning-loop}
\end{figure}
```

9 Conclusion

The results of this work show that using AI for parsing foreign ERP exports into a unified DTO schema is not only feasible but also strategically justified. Hand-crafted, vendor-specific parsers quickly become brittle as schemas evolve, tenants customize fields, and new systems are added; in contrast, a layered AI pipeline can absorb much of this variability without requiring exhaustive rule maintenance. By combining deterministic mappings with fuzzy normalization and an LLM fallback, the connector can reliably interpret

63. **techtarget2024ruleml**; Gunning, "Explainable Artificial Intelligence (XAI)."
64. Dataforest, "Enterprise Data Integration Strategy & Architecture."

heterogeneous field names, value encodings, and record structures and still emit consistent, well-typed DTOs.

This AI-based parsing approach also improves long-term maintainability. Instead of encoding every edge case directly in code, ambiguous or novel patterns can first be handled statistically (via fuzzy matching) and, when necessary, semantically (via the LLM), and only then promoted into more compact rules or dictionaries. Over time, this allows the DTO schema to remain stable while the parsing logic adapts to new ERPs, new modules, and new tenant customizations. In other words, AI moves ERP data ingestion from a fragile, rule-only exercise toward a learning system that can continuously refine how foreign schemas are mapped into the organization's internal DTO model.“‘

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